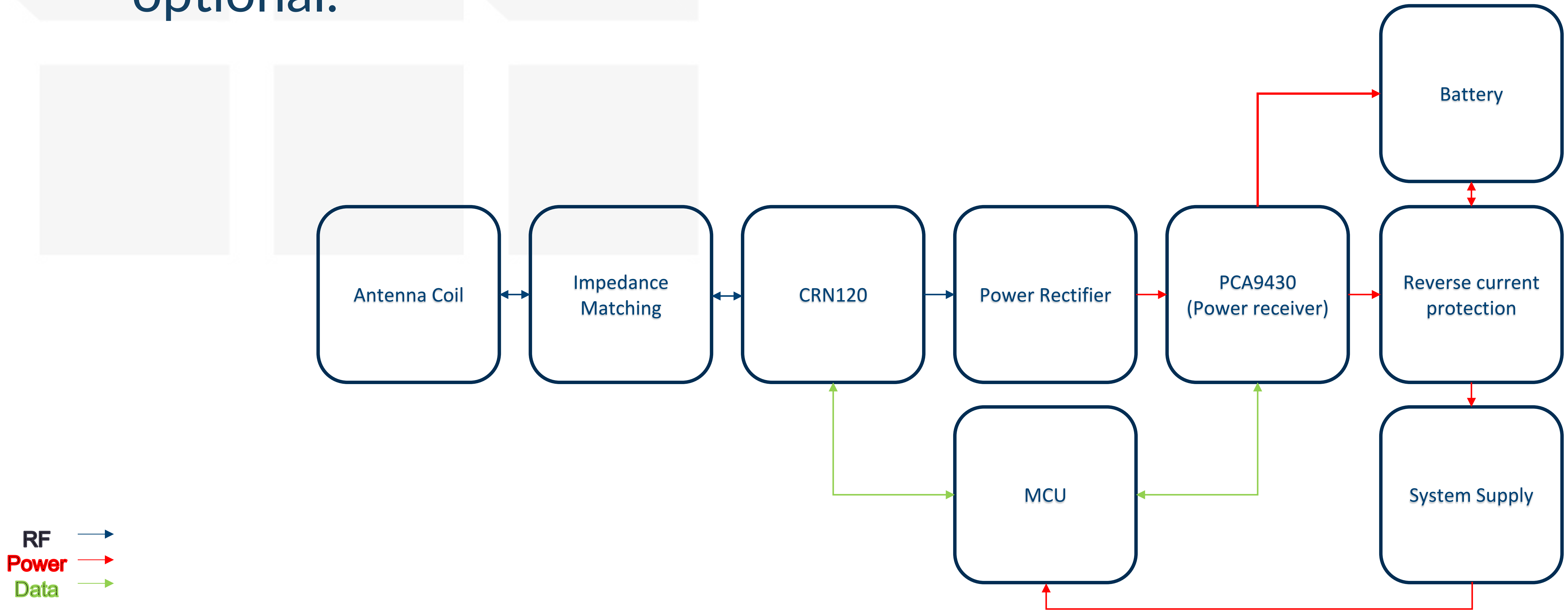
Must have components

- The simplified block diagram of must have components for enabling the WLC system designed by CISC for Sibel is shown bellow.
- Any other components (like EEPROM, Leds, Sensors, etc.) are considered as optional.



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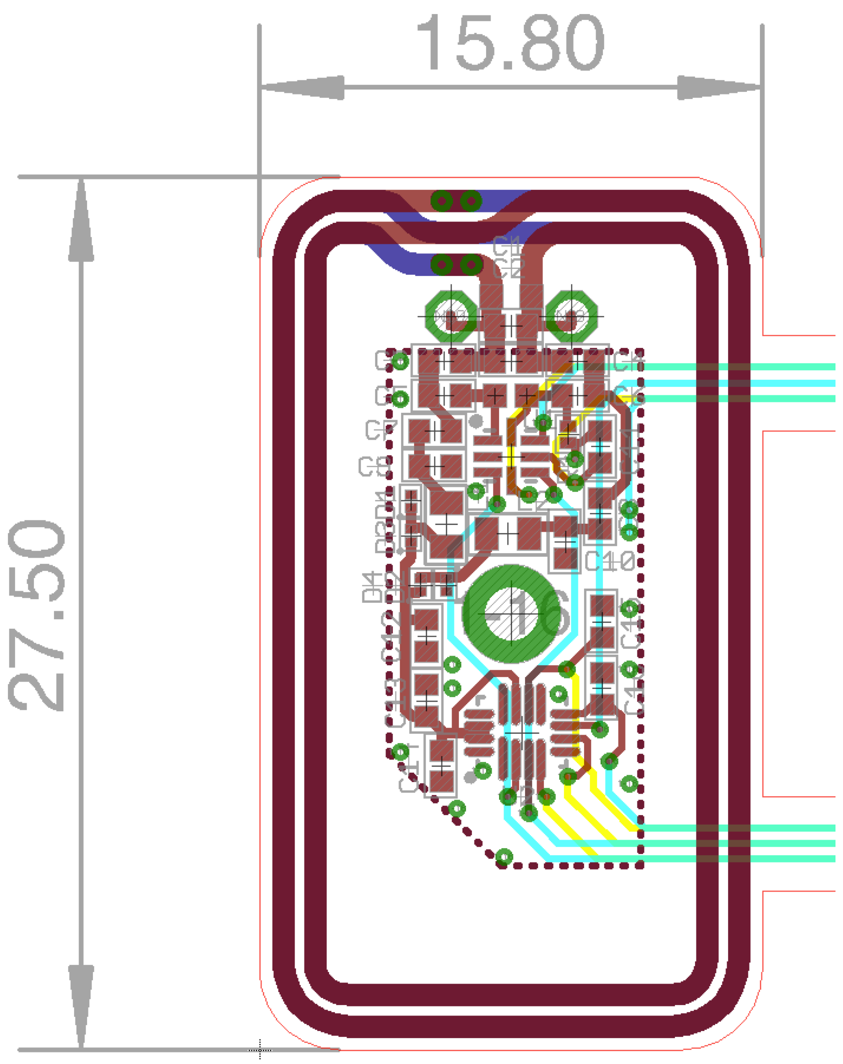
Antenna

- Customized antenna has been designed:



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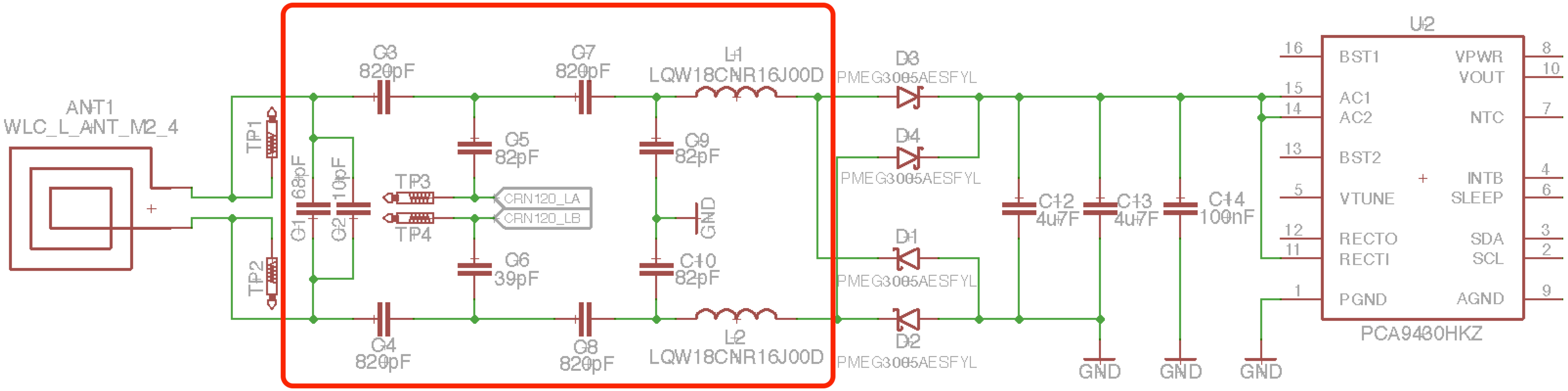
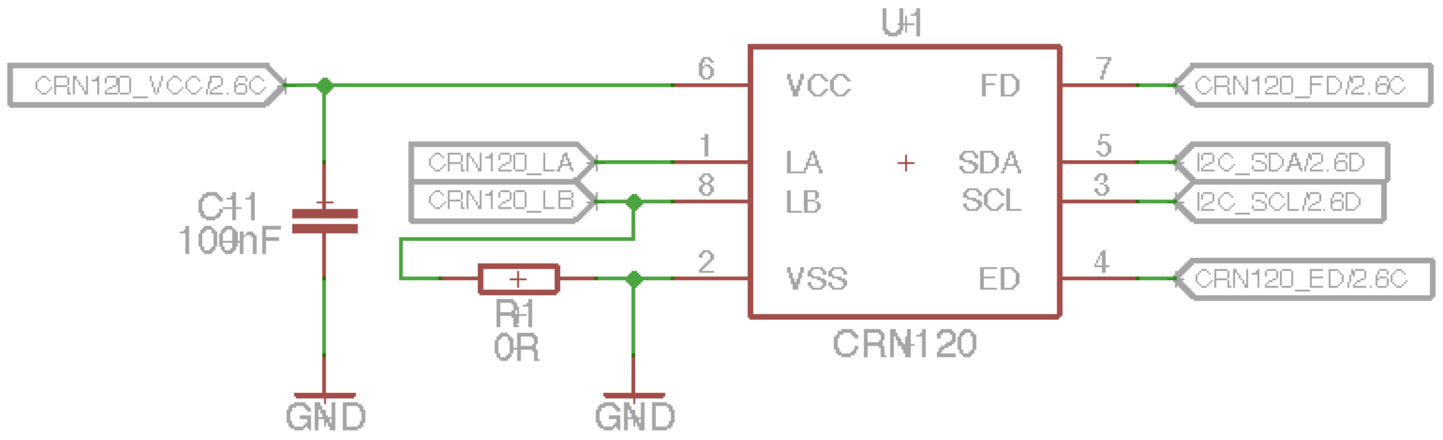
Impedance Matching

- The designed matching structure together with the value of the components:



CRN120

- The design of the CRN120, which is the NFC wireless charging communication receiver frontend:
  - ❖ CRN120_VCC has range of 1.67 – 3.6V. It's recommended to have option to enable/disable the CRN120_VCC supply voltage, serving as a reset.
  - ❖ CRN120_LA and CRN120_LB are connected to the Impedance Matching structure.
  - ❖ CRN120_FD is an optional pin, recommended to be connected to MCU GPIO Input pin.
  - ❖ I2C_SDA and I2C_SCL are I2C communication lines (up to 400kHz).
  - ❖ CRN120_ED is a must-have pin connected to the MCU GPIO Input pin. An external pull-up is recommended, but internal MCU Pull up can be used if needed.



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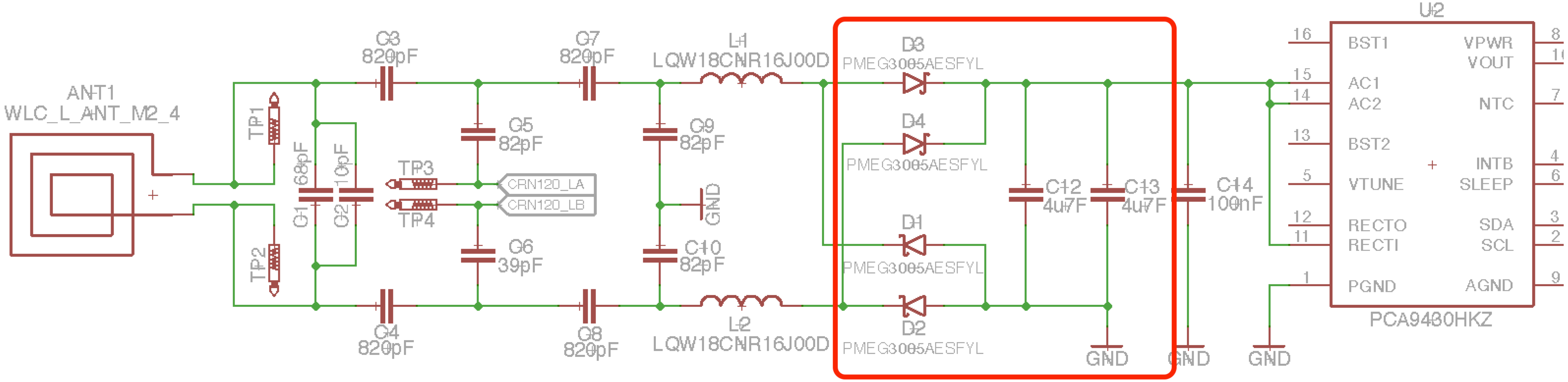
Power Rectifier

- The impedance matching and power measurements has been done with this components used for power rectifier:



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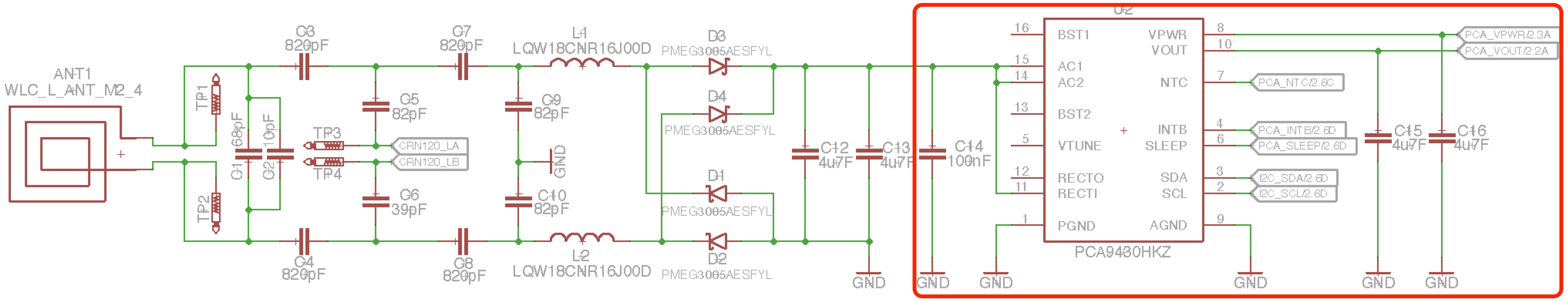
Power Rreceiver

- PCA9430 is used as the NFC charging power receiver with linear battery charger:
  - ❖ PCA_VPWR serves as a dead-battery LDO.
  - ❖ PCA_VOUT is the output of the linear battery charger.
  - ❖ PCA_NTC is an optional pin and can be used as external temperature sensor
  - ❖ PCA_INTB is a must-have pin, connected to the MCU GPIO Input. An external pull-up is recommended, but internal MCU Pull up can be used if needed.
  - ❖ PCA_SLEEP is an optional pin recommended to be connected to MCU GPIO Input pin.
  - ❖ I2C_SDA and I2C_SCL are I2C communication lines (up to 1 MHz, but it's recommended to use same clock as for CRN120).



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Battery

- Any type of a Lithion-Ion polymer Battery can be used.
- CELLEVIA BATTERIES LP401230 is used for the CISC WDK SensorTag.

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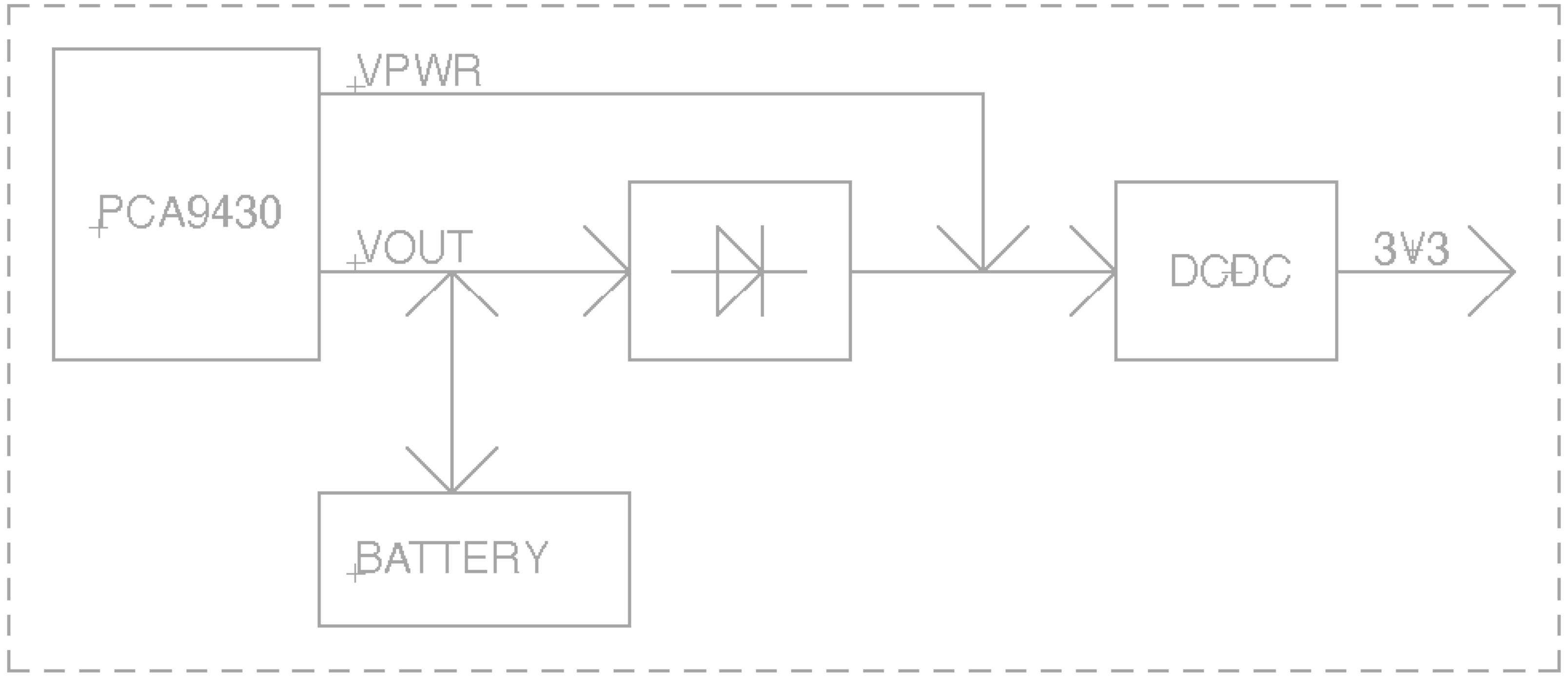
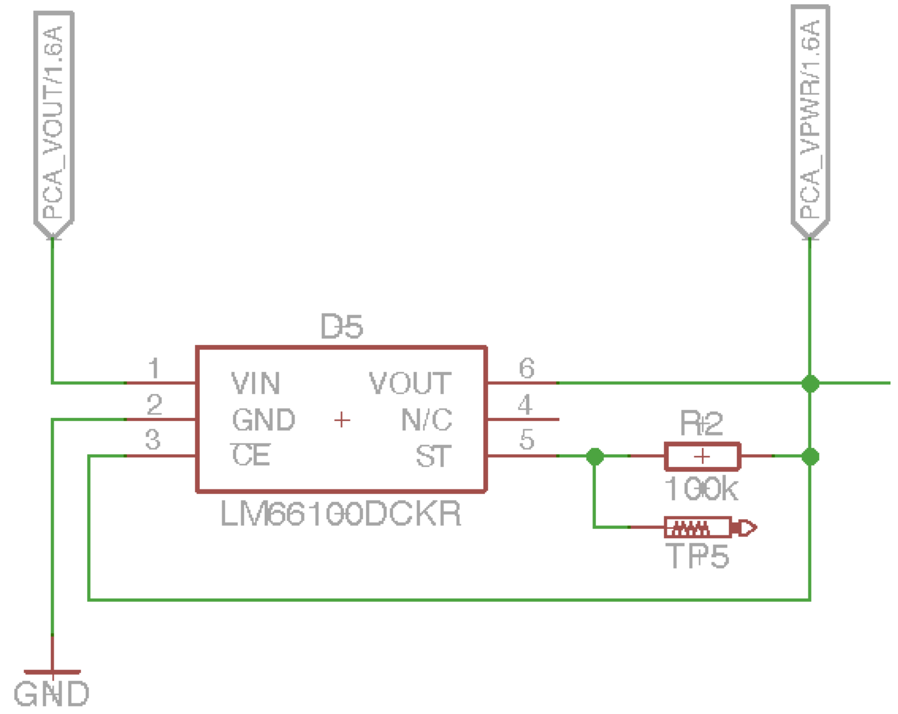
Reverse current protection

- In order to be able to interconnect the Battery with the Dead-Battery LDO of the Power receiver, it's necessary to implement a reverse current protection circuit.
- Any type of reverse current protection design can be used (Diode, LDO with reverse current protection, etc.)
- Current design is based on an ideal diode: LM66100DCKR





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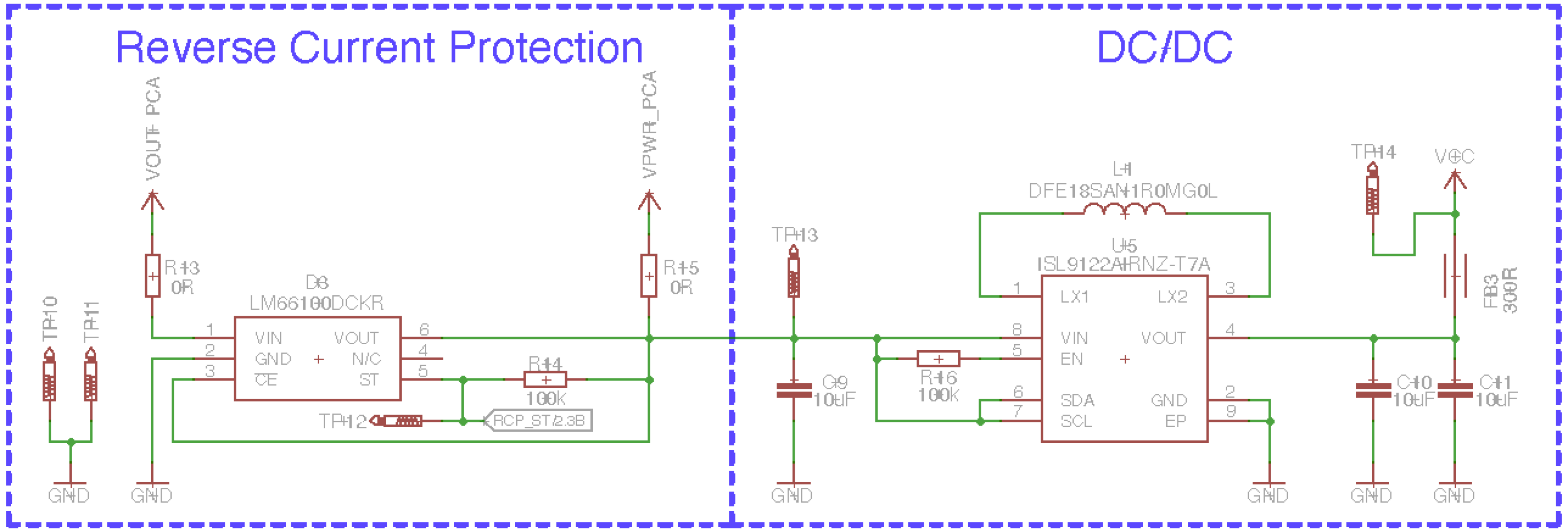
System supply

- This is theoretically an optional part, but if the design should not use directly the battery to supply the system, an additional DC/DC (or LDO) is required.
- Any type of a DC/DC (or LDO) can be used.
- System supply is connected directly to the output of the Reverse Current Protection.
- Example of a connection of the DC/DC in the CISC WDK SensorTag:



MCU

- MCU must run a WLC Device firmware.
- Theoretically any type of a MCU can be used, since the WLC Device Library is portable.
- For Sibel, WLC Device firmware has been ported for the Nordic nRF52840 MCU, running the Zephyr OS.

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